

LibSQL TLS Error - FIXED

The Error

```
thread 'main' panicked at libsql-0.9.29/src/database.rs:783:5:  
The `tls` feature is disabled, you must provide your own http connector
```

Root Cause

The `libsql` crate needs TLS features enabled to connect to Turso Cloud (which uses HTTPS).

Solution Applied

File: `backend/Cargo.toml`

Changed from:

```
libsql = "0.6"
```

Changed to:

```
libsql = { version = "0.6", features = ["core", "replication", "sync"] }
```

How to Apply

Option 1: Use Updated Archive

Download the latest `blog-system-FINAL-WITH-DEPLOY.tar.gz` which has this fix.

Option 2: Manual Update

```
cd backend
```

```
# Edit Cargo.toml and change the libsql line to:
```

```
# libsql = { version = "0.6", features = ["core", "replication", "sync"] }
```

```
# Clean rebuild
```

```
cargo clean
```

```
cargo build
```

```
# Deploy
```

```
fly deploy
```

Verification

After fixing, the build should succeed:

```
cd backend
cargo build --release
```

Expected:

```
Compiling libsql v0.6...
Compiling blog-backend v0.1.0
Finished release [optimized] target(s)
```

No panic!

What Each Feature Does

Feature	Purpose
core	Basic database operations
replication	Turso replication support
sync	Synchronization features

These features enable the HTTPS/TLS connector needed for Turso Cloud.

Alternative: If Still Fails

If you still see TLS errors, try the full feature set:

```
libsql = { version = "0.6", features = ["core", "replication", "sync", "encryption"] }
```

Or use all features:

```
libsql = { version = "0.6", features = ["full"] }
```

Common Related Issues

Issue 1: “rustls” missing

Error:

```
error: rustls not found
```

Fix: Add to Cargo.toml:

```
rustls = "0.21"
```

Issue 2: “openssl” missing

Error:

Could not find openssl

Fix (Dockerfile):

```
# Add to your Dockerfile before COPY
RUN apt-get update && apt-get install -y \
    pkg-config \
    libssl-dev \
    && rm -rf /var/lib/apt/lists/*
```

Issue 3: Timeout connecting

Error:

Connection timeout

Fix: Check your Turso credentials:

```
# Verify database exists
turso db list

# Test connection
turso db shell blog-production
```

Complete Deploy Commands

After fixing Cargo.toml:

```
cd backend

# Clean build (important!)
cargo clean

# Test locally first
cargo build

# If successful, deploy to Fly.io
fly deploy

# Or rebuild Docker image
docker build -t blog-backend .
```

Updated Cargo.toml

Here's the complete corrected dependencies section:

```
[dependencies]
tokio = { version = "1.42", features = ["full"] }
axum = "0.8"
tower = "0.5"
tower-http = { version = "0.6", features = ["cors", "trace", "fs"] }
serde = { version = "1.0", features = ["derive"] }
serde_json = "1.0"
libsql = { version = "0.6", features = ["core", "replication", "sync"] } # ← Fixed!
chrono = { version = "0.4", features = ["serde"] }
uuid = { version = "1.11", features = ["v4", "serde"] }
bcrypt = "0.16"
jsonwebtoken = { version = "9.3", features = ["rust_crypto"] }
tracing = "0.1"
tracing-subscriber = "0.3"
dotenvy = "0.15"
base64 = "0.22"
image = "0.25"
webp = "0.3"
```

Testing After Fix

Local Test

```
cd backend

# Set env vars
export TURSO_DATABASE_URL="libsql://your-db.turso.io"
export TURSO_AUTH_TOKEN="your-token"
export JWT_SECRET="test-secret"

# Run
cargo run

# Should see:
# Server running on http://0.0.0.0:3001
```

Fly.io Test

```
# Deploy
fly deploy

# Check logs
```

```
fly logs
```

```
# Should NOT see TLS error  
# Should see: Server running...
```

```
# Test endpoint  
curl https://your-app.fly.dev/health  
# Should return: {"status":"ok"}
```

Why This Happens

LibSQL versions: - `libsql = "0.6"` → Uses **default features** (no TLS) -
`libsql = { version = "0.6", features = [...] }` → Enables TLS

Turso requires TLS because it's a cloud service using HTTPS.

Local SQLite doesn't need TLS because it's a file.

Pro Tip: Check Features

To see what features a crate supports:

```
# Check libsql features  
cargo tree -f "{p} {f}"  
  
# Or check on crates.io  
open https://crates.io/crates/libsql
```

Summary

The Fix: 1. Add features to `libsql` in `Cargo.toml` 2. Rebuild with `cargo clean && cargo build` 3. Deploy with `fly deploy`

Result: - TLS support enabled - Can connect to Turso - No more panic errors - Production ready!

Related Docs

- LibSQL features: <https://docs.rs/libsql/latest/libsql/>
 - Turso Rust SDK: <https://docs.turso.tech/sdk/rust/quickstart>
 - Fly.io Rust: <https://fly.io/docs/languages-and-frameworks/rust/>
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Fixed! Your backend will now connect to Turso correctly.